

## SI system

- SI stands for *Système Internationale d'Unités*.
- It is a global standard of measurement used by scientists so results can be shared, compared, and repeated anywhere.

## SI UNIT BASE CHART

| Quantity           | SI Unit  | US Unit                    | Symbol       | Conversion                                      | Measuring tool             |
|--------------------|----------|----------------------------|--------------|-------------------------------------------------|----------------------------|
| <b>Length</b>      | Meter    | inch, foot                 | m            | 1 inch $\approx$ 2.54 cm                        | Ruler, Measuring tape      |
| <b>Mass</b>        | Kilogram | pound (lb)                 | kg           | 1 lb $\approx$ 0.45 kg                          | Balance                    |
| <b>Volume</b>      | Liter    | gallon/ounce               | L            | 1 gallon $\approx$ 3.78 liters                  | Beaker, Graduated cylinder |
| <b>Temperature</b> | Celcius  | Fahrenheit ( $^{\circ}$ F) | $^{\circ}$ C | $(^{\circ}$ F - 32) $\times$ 5/9 = $^{\circ}$ C | Thermometer                |
| <b>Time</b>        | Second   | second, minute             | sec          | (same in both systems)                          | Stopwatch, Clock           |

## Part A: Match the Quantity with the SI Unit

Draw a line to match.

### PHYSICAL QUANTITY

### SI UNIT

Length

Kilogram

Mass

Second

Time

Metre

Temperature

Ampere

Electric Current

Centigrade/Kelvin/Fahrenheit

## Part B: Fill in the Missing Unit

Choose from:

**metre (m), kilogram (kg), second (s), kelvin (K), ampere (A)**

1. Length of a classroom = \_\_\_\_\_
2. Mass of a watermelon = \_\_\_\_\_
3. Time taken to complete a race = \_\_\_\_\_
4. Temperature of a substance = \_\_\_\_\_
5. Electric current in a circuit = \_\_\_\_\_

## Part C: Write the Correct Symbol

| SI UNIT  | SYMBOL |
|----------|--------|
| Metre    | _____  |
| Kilogram | _____  |
| Second   | _____  |
| Length   | _____  |

## Part D: Think Like a Scientist

Identify the quantity being measured.

### Example:

A student measures the height of a plant.

**Quantity:** Length

**SI Unit:** metre (m)

1. A student measures the mass of a science textbook.

Quantity: \_\_\_\_\_

SI Unit: \_\_\_\_\_

2. A student records how long sugar takes to dissolve.

Quantity: \_\_\_\_\_

SI Unit: \_\_\_\_\_

3. A scientist measures the temperature of water.

Quantity: \_\_\_\_\_

SI Unit: \_\_\_\_\_

4. An electrician measures current flowing through a wire.

Quantity: \_\_\_\_\_

SI Unit: \_\_\_\_\_

### Part E : Real-Life Applications

Write the most suitable SI unit.

1. Length of a football field = \_\_\_\_\_
2. Mass of your school bag = \_\_\_\_\_
3. Time taken to blink = \_\_\_\_\_
4. Temperature of boiling water = \_\_\_\_\_
5. Current in a battery circuit = \_\_\_\_\_

### Part F : Convert the Units

Convert the following into **SI Units**:

1. 12 inches = \_\_\_\_\_ cm
2. 3 gallons = \_\_\_\_\_ liters
3.  $98.6^{\circ}\text{F}$  = \_\_\_\_\_  $^{\circ}\text{C}$
4. 5 pounds = \_\_\_\_\_ kilograms
5. 60 minutes = \_\_\_\_\_ seconds

**Reflection Prompt:**

Why is it important for scientists around the world to use the same system of units?

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